MIL-D-50720(MU)
10 May 1971
SUPERSEDING
LCA-PD-59
16 April 1970

MILITARY SPECIFICATION

# DUMMY CARTRIDGE, 20 MILLIMETER: PLASTIC, XM254

## 1. SCOPE

1.1 This specification covers one type of 20mm plastic dummy cartridge, with a steel head and stem for body reinforcement, to be used in the 20mm M39 and M61 type automatic gun systems.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-171 | - Finishing of Metal and Wood Surfaces |
| MIL-STD-644 | - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition |
| MIL-STD-651 | - Visual Inspection Standards for 20mm Ammunition and Components |
| MIL-STD-1168 | - Lot Numbering of Ammunition |
| MIL-STD-1235 | - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes |

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

| | |
|---|---|
| F7258942 | - Method of Packing 200 Cartridges, 20mm, with M103 Type Case, in Bulk, in M548 Container |
| D7259079 | - Marking on Shipping and Storage Container, M548, for Cartridges, 20mm, Dummy |
| D7259254 | - Dummy Cartridge, 20mm: Plastic, XM254 |

FSC 1305

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2.1 DRAWINGS (cont)

IEL 7259254 - Index of Inspection Equipment Lists for: Dummy Cartridge, 20mm: Plastic, XM254
D7259296 - Head and Stem Assembly

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505 - Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)
ASTM A219 - Standard Methods of Test for Local Thickness of Electrodeposited Coatings (Appendix 1)

(Application for copies should be addressed to: American Society for Testing Materials, 1916 Race Street, Philadelphia, Pa. 19137)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They also are distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General.-The cartridge shall comply with the requirements of Drawing (Dwg.) D7259254, referenced specifications and other requirements specified herein.

3.2 Manufacturing process.-All parts and assemblies shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without his prior approval. (see 6.2)

3.3 Protective coating.-The protective finish of the head shall be as specified on the applicable drawing. The coating shall have a uniform appearance, shall be adherent and free from blisters and substantially free from other defects that may affect the appearance or protective value of the coating.

3.3.1 Protective coating thickness.-The thickness of the finished coating on the cartridge head shall be not less than 0.0002 inch.



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3.4 Plastic security.-The cartridge shall withstand a tension load as specified on Dwg. D7259254, without separation or breakup of the plastic body from the steel head or breakup of the plastic portion.

3.5 Workmanship.-The requirements for workmanship are as specified by applicable drawings, referenced specifications and the following:

3.5.1 Processing defects.-The cartridge shall be free from cracks, splits, fins, flashing, deformities, porosity and other forms of substandard workmanship.

3.5.2 Cleanliness.-The cartridge shall be free from all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or other facilities suitable for the performance of the inspection requirements specified therein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of:

a. For visual and dimensional inspection:

50 Stems
50 Heads
50 Head and Stem Assemblies
100 Cartridges

b. For tests:

52 Cartridges



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4.2.1 (cont)

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. Where multiple cavity molds are used the sample cartridges shall be identified as to mold cavity. Each mold cavity shall be equally represented and identified in the sample. The cartridges shall be identified in accordance with MIL-STD-1168.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, as directed by the contracting officer, the sample will be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified by the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.1 Lot.-A lot shall consist of dummy cartridges produced by one manufacturer in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

4.3.1.2 Lot identification.-Each dummy cartridge and each packed cartridge lot shall be identified in accordance with applicable drawings and MIL-STD-1168. Each packed lot shall be further identified by a federal stock number assigned by the procuring agency.

4.3.2 Examination.-Examination for major and minor defects shall be performed as specified herein. Each component, assembly or cartridge found to be defective shall be rejected.

4.3.2.1 Sampling and acceptance criteria.-Sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. Defect characteristics may be grouped into a class and considered collectively, or defect characteristics may be considered individually to determine acceptability. When acceptability is determined on a class basis, in addition to the Acceptable Quality Level (AQL) assigned to the major class, an AQL shall be assigned to individual characteristics within the major class, limiting the number of defectives containing a particular major defect to less than the number of defectives permitted for the class. However, where MIL-STD-1235 is used, sampling plans applicable to a class shall not be used for major defectives. (see 6.4)



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4.3.2.1.1 AQL's.-The following AQL's are assigned to major and minor defects:

a. Individual:

Major ---------- 0.25%
Minor ---------- 0.40%

b. Class basis:

Major ---------- 1.50%
Minor ---------- 2.50%

c. Individual defects associated with the class basis:

Major ---------- 0.40%

4.3.3 Classification of defects 1/.-The classification of defects shall be as follows:

4.3.3.1 Head (see Dwg. C7259255, a detail of Dwg. D7259296).

| Categories and Defects | Method of Inspection |
|---|---|
| **Critical:** None defined. | |
| **Major:** | |
| 101. Diameter, datum surface C | Gage |
| 102. Concentricity, head rim, with datum surface C | Gage |
| 103. Spherical radius missing, head face | Visual |
| 104. Foreign matter (including corrosion) | Visual |
| 105. Diameter of major cavity | Gage |
| **Minor:** | |
| *201. Diameter, reference surface B | Gage |
| *202. Diameter, undercut, in reference surface B | Gage |
| *203. Width, undercut, in reference surface B | Gage |
| *204. Length, head, to end of datum surface C,min. | Gage |
| 205. Surface finish, above datum surface C | Visual |

4.3.3.2 Stem (see Dwg. C7259297, a detail of Dwg. D7259296).

| Categories and Defects | Method of Inspection |
|---|---|
| **Critical:** None defined. | |
| **Major:** | |
| 101. Diameter of stem | Gage |
| **Minor:** None defined. | |



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4.3.3.3 Head and stem assembly (see Dwg. D7259296).

| Categories and Defects | Method of Inspection |
|---|---|
| **Critical:** None defined. | |
| **Major:** None defined. | |
| **Minor:** | |
| 201. Concentricity, stem with datum surface | Gage |
| 202. Length, max. | Gage |

4.3.3.4 Dummy cartridge (see Dwg. D7259254 or C7259255, a detail of Dwg. D7259254, as applicable).

| Categories and Defects | Method of Inspection |
|---|---|
| **Critical:** None defined. | |
| **Major:** | |
| 101. Thickness, head rim, min. | Gage |
| 102. Thickness, head rim, max., and perpendicularity with datum surface A | Gage |
| 103. Diameter, head rim, min. | Gage |
| 104. Diameter, extractor groove, max. | Gage |
| 105. Width, extractor groove, min. | Gage |
| 106. Concentricity, head rim, with datum surface A | Gage |
| **107. Length to .942 datum reference surface | Gage |
| **108. Length, to leading edge of rotating band, max. | Gage |
| 109. Profile, crimp groove | Gage |
| 110. Length, to centerline of crimp groove | Gage |
| **111. Diameter, bourrelet, max. | Gage |
| **112. Diameter, rotating band, max. | Gage |
| **113. Diameter, at 1.750 datum | Gage |
| 114. Total length, max. | Gage |
| 115. Profile and alignment, max. | Gage |
| 116. Chamfer missing, head rim | Visual |
| 117. Protective coating missing | Visual |
| 118. Step or gap, junction of head and body | Visual |
| 119. Finish, plastic portion | Visual 2/ |
| 120. Foreign matter (including steel corrosion) | Visual |
| 121. Weight | Scale |
| **Minor:** | |
| 201. Diameter, head rim, max. | Gage |
| 202. Length of flat on head rim | Gage |
| 203. Diameter, extractor groove, min. | Gage |
| **204. Diameter, bourrelet, min. | Gage |



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4.3.3.4 (cont)

| Categories and Defects | Method of Inspection |
|---|---|
| Minor: (cont) | |
| **205. Diameter, rotating band, min. | Gage |
| 206. Width, extractor groove, max. | Gage |
| **207. Total length, min. | Gage |
| 208. Depth, spherical radius, min. | Gage |
| 209. Protective coating defective | Visual 3/ |
| 210. Damage, except to protective coating | Visual 4/ |
| 211. Nib on projectile nose | Visual |
| 212. Marking incorrect, incomplete, illegible or missing | Visual |

*Upon prior approval by the contracting officer, inspection for defect characteristics identified by a single asterisk may be reduced after three lots have been accepted, to 1 in 5 production lots, provided that the inspection results of each sample of three consecutive lots have complied with all applicable acceptance criteria of the defect characteristic so identified.

**Upon prior approval by the contracting officer, inspection for defect characteristics identified by double asterisks may be reduced after three lots have been accepted, to 1 in 5 production lots.

1/ MIL-STD-651 shall apply in defining and evaluating visual defects of the cartridge head to the extent covered.

2/ A crack, split, chip, air bubble or other visible form of porosity in the plastic body shall be cause for classifying the cartridge defective.

3/ Nodules, scales, blisters, burns or bare spots in the plating of the head shall be cause for classifying the cartridge defective.

4/ Protective coating which is damaged to expose basis metal, a deep cut, dent or other visual defect which is indicative of poor workmanship or careless material handling, or which presents a hazard to weapon cycling, shall be classed defective.

4.3.4 Tests.-The tests listed in Table I shall be performed on each lot in accordance with the test methods prescribed by 4.4. For acceptance, each test shall comply with the applicable requirement(s).

4.3.4.1 The sampling plans for the various tests shall be as specified by Table I.



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4.3.4.1 (cont)

Table I

| Test | Sample Size | Item | Rqmt Para |
|---|---|---|---|
| Protective coating thickness | 1/ 32 | Cartridges | 3.3.1 |
| Plastic security | 2/ 20 | Cartridges | 3.4 |

1/ Failure of three or more sample cartridges to comply with the minimum coating thickness requirement shall be cause for rejection of the lot. If two or less cartridges fail to comply with the minimum thickness requirement, a second sample consisting of 32 cartridges shall be tested. If, in the accumulated samples, four or more cartridges fail to comply with the minimum thickness requirement, the lot shall be rejected.

2/ Separation of the plastic from the steel head or breakup of the plastic portion of one or more cartridges of the sample as a result of the specified tension load for plastic security, shall be cause for rejection of the lot.

4.3.5 Packaging, packing and marking inspection.-Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644, as applicable.

4.3.6 Inspection equipment.-Index of Inspection Equipment List No. IEL 7259254 identifies the applicable inspection equipment list required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL (reference is made to inspection equipment list for definition of coding format).

4.4 Test methods and procedures.

4.4.1 Protective coating thickness.-The magnetic method as specified by ASTM A219, Appendix 1, shall apply in the determination of coating thickness. Two measurements shall be taken on opposite flat surfaces of the head face of each sample cartridge. The average of the two measurements shall apply as the coating thickness.

4.4.2 Plastic security.-The equipment and method used to determine plastic security shall comply with applicable requirements of AMCR 715-505, Volume 8, for projectile extraction.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level A.-Each lot of cartridges shall be packed as specified by Dwg. F7258942.

5.2 Marking.-Marking of containers shall comply with applicable requirements of Dwg. D7259079.



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6. NOTES

6.1 Ordering data.-Invitation for bids and contracts or orders should specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Place of inspection if not place of manufacture.

6.1.3 Provision for supply of Government furnished shipping and storage containers, M548. (see 5.1)

6.1.4 Detailed packing and marking instructions. (see 5.1 and 5.2)

6.1.5 Provision for the supply, maintenance and disposition of mandatory equipment for acceptance inspection purposes.

6.1.6 Provision for submission and approval of the manufacturing process.

6.2 Process deviation.-A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item.

6.3 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

Custodian:

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Preparing Activity:

Army - MU

Project No. 1305-A669



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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